AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129

Technologies in Practice

Final Examination
Spring Semester 2025/2026

Date: 11/Jun/2026

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Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 h

Instructions:

- Total Marks: 50
- Total Number of Questions:
  - Question 1: 10
  - Question 2: 3
  - Question 3: 2
  - Question 4: 3
  - Question 5: 4
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

Question 1: Answer the following questions [10 Marks]

1. The amount of time the schedule takes from when a request was submitted until the first response is produced is known as.
   a. Waiting time
   b. Response time
   c. Turnaround time
   d. Throughput
2. In deadlock, a condition where only one thread at a time can use the resource is called?
   a. No preemption
   b. Hold and wait
   c. Mutual exclusion
   d. Circular wait
3. A way to determine robot motion while using a Cartesian-based frame, with the thumb, forefinger, and middle finger?
   a. Center-point rule
   b. Left-hand rule
   c. Right-hand rule
   d. Frame
4. What is the purpose of the "Step Mode" during program testing?
   a. Runs the program at full speed
   b. Deletes errors automatically
   c. Saves the program automatically
   d. Executes the program one line at a time
5. What is the main purpose of subroutines in robot programming?
   a. To increase program complexity
   b. To reduce lines of code and simplify programming
   c. To eliminate motion instructions
   d. To replace the main program
6. Which IEEE standard is used to represent Bluetooth?
   a. IEEE 802.2
   b. IEEE 802.3
   c. IEEE 802.11
   d. IEEE 802.15
7. Which network type typically spans only a few meters around an individual and is commonly used in IoT?
   a. PAN
   b. LAN
   c. MAN
   d. WAN
8. Which field does **NOT** commonly rely on robotic systems for its core operations?
   a. Automotive manufacturing
   b. Online banking services
   c. Space exploration
   d. Minimally invasive surgery
9. An industrial robot is defined by the Robotic Industries Association RIA as_____?
   a. A reprogrammable, multipurpose manipulator
   b. A machine that replaces all human labor
   c. A fixed mechanical device
   d. A mobile AI system only
10. The directed edge ($T_i \rightarrow R_j$) from a thread T_i to Resource R_j is known as
    a. Assignment edge
    b. Allocation
    c. Request edge
    d. Maximum

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Question 2: Answer the following questions [10 Marks]

a) Explain the process of context switch when CPU switches from one process to another? (2 Marks)
b) Explain with diagram the process states? (2 Marks)
c) Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process using: -
    1) First Come First Serve (3 Marks)
    2) Preemptive Shortest Job First (3 Marks)

Process	P1	P2	P3	P4	P5
Burst time	7	4	12	6	9
Arrival time	1	3	2	0	5

Question 3: Answer the following questions [10 Marks]

a) Describe the role of sensors and actuators in an IoT-based automation system? Explain how sensors and actuators work together to support automation in smart greenhouse system. (4 Marks)

b) Suppose we have two resources, A, and B. A has 8 instances and B has 6 instances. Can the system execute the following processes without deadlock occurring? Show the process sequence with detailed calculations. (6 Marks).

Process	Allocate		Maximum need	
	A	B	A	B
P1	1	2	5	3
P2	2	1	4	4
P3	1	1	2	2
P4	1	0	6	1

Question 4: Answer the following questions [10 Marks]

a) Match the following components of the Cisco IoT System with their roles. (4 Marks)

1. Edge Devices	a) Ensures encrypted communication and device authentication in IoT environments.
2. Cloud Computing	b) Physical objects such as sensors and actuators that collect and transmit data.
3. IoT Gateway	c) Provides large-scale storage and computational power for big data processing.
4. Encryption Protocols	d) Acts as an intermediary that aggregates data and connects local devices to the internet.

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b) How do the 802.11n and 802.11ac Wi-Fi standards differ in terms of speed, frequency, and compatibility? (4 Marks)

c) Differentiate between Open loop control systems and closed loop control systems in IoT. (3 Marks)

Question 5: Answer the following questions [10 Marks]

a) A metropolitan city is deploying IoT-enabled traffic sensors, smart traffic lights, and connected surveillance cameras to reduce congestion and improve road safety. How does Cisco's IoT System architecture support real-time traffic monitoring, device connectivity, secure data transmission, and automated traffic signal control to enhance urban mobility? (3 Marks)

b) Rank the following IoT communication protocols from lowest to highest in terms of power consumption:(2 Marks)
   1) Zigbee
   2) Cellular (3G/4G/5G)
   3) NFC (Near Field Communication)
   4) LoRaWAN

c) You are designing a smart grid system that enables utility providers to monitor electricity consumption, detect faults, and remotely manage smart meters through a centralized control platform. Which IoT component enables reliable communication between smart meters, grid sensors, and the central management system to ensure real-time monitoring and control? (2 Marks).

d) Explain about the three main zones around the robot in terms of safety? (3 Marks)

End of Questions

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